Exploiting Locality for Efficient Conversational Search

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2018 ····· **2018-2019** ····· **2019** ····· **2020** ····· **2021** ····· **2022** ····· **2023 →**

2018

Search is no longer a single-shot query

"Towards Conversational Search and Recommendation" [Zhang et al.], QuAC [Choi et al.]

2018-2019

The Transformer Era Changes Everything

BERT [Devlin et al.]
Passage Re-ranking with BERT [Nogueira & Cho]

2019

Conversational search became *measurable*

TREC CAsT 2019
Query rewriting→
BM25 retrieve →
BERT re-rank

2020

Dense Retrieval Emerges

DPR [Karpukhin et al]
ColBERT[Khattab & Zaharia]
RAG [Lewis et al.]

2021

Conversational query rewriting improves

ConvDR [Yu et al.]
Contextualized query embeddings

2022

LLMs Enter the Scene

RAG from research system to infrastructure.

Dense retrieval is now everywhere.

2023 →

LLM-native conversational search

Perplexity, Bing Chat, ChatGPT, Claude
Generative AI
Agentic RAG

Efficiency in conversational search

Efficiency and scalability

- Retrieval effectiveness has improved dramatically with transformers, dense models, and LLMs, but measurable gains are now slowing.
- Latency and computational costs are rising. Scalability becomes the bottleneck.
- Efficiency is no longer an implementation detail—it is a core research challenge.

Efficiency and scalability

If we solve conversational search quality, is the system *fast enough* for interactive use?

- Retrieval effectiveness has improved dramatically with transformers, dense models, and LLMs, but measurable gains are now slowing.
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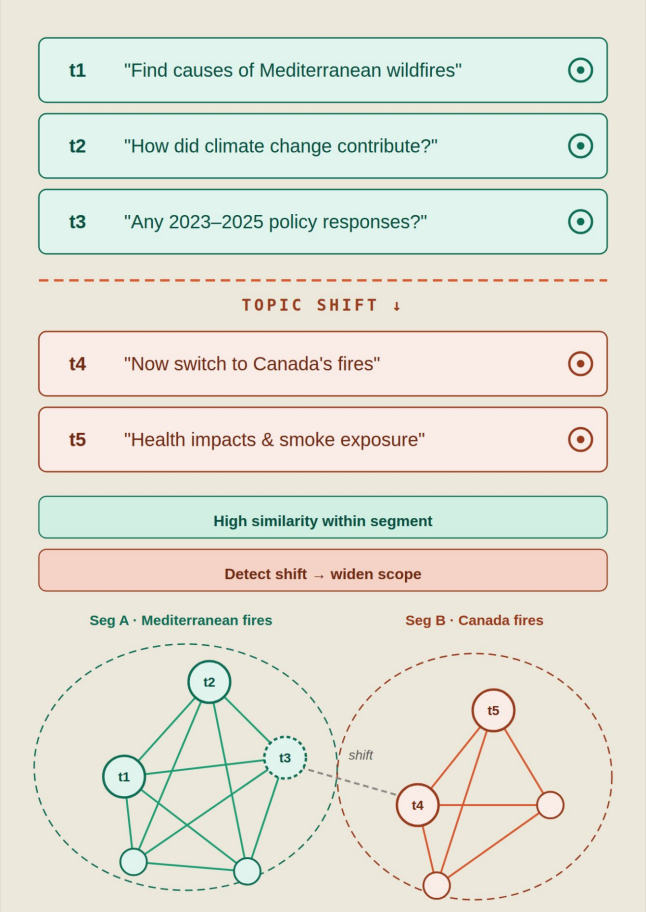
Topical locality



→ Restrict the search space dynamically

→ Cache and reuse previously retrieved answers

→ Restrict search to relevant subspaces

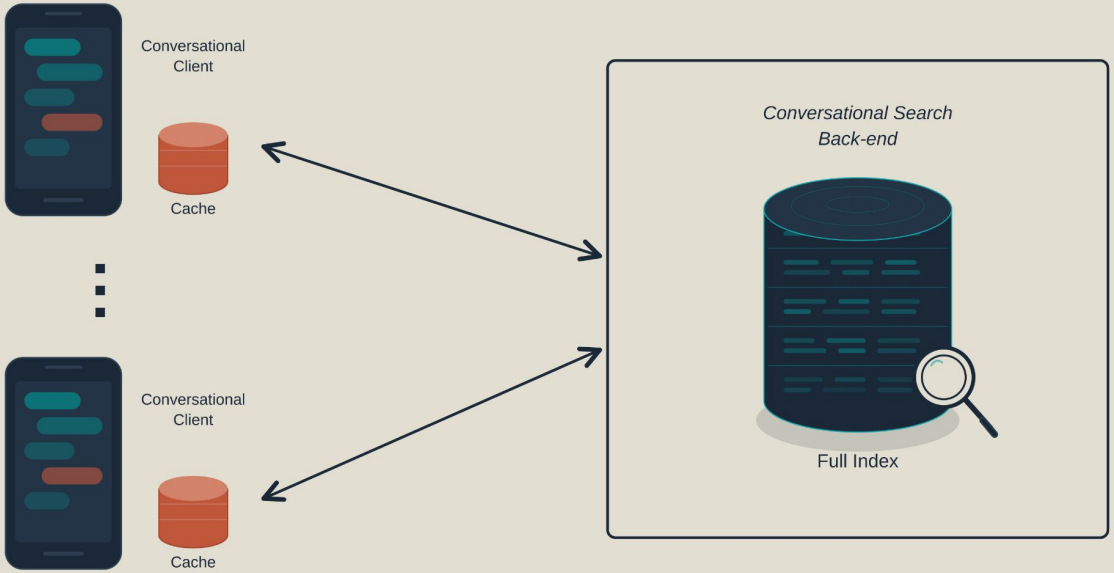
→ Avoid search from scratch for follow-up turns

Experimental Setting:

★ Dense retrieval, ANN search
★ TREC CAsT 2019-2021

Search locally, not globally

Temporal locality



Spatial locality

Cache document embeddings on the client; reuse for subsequent turns via metric space properties

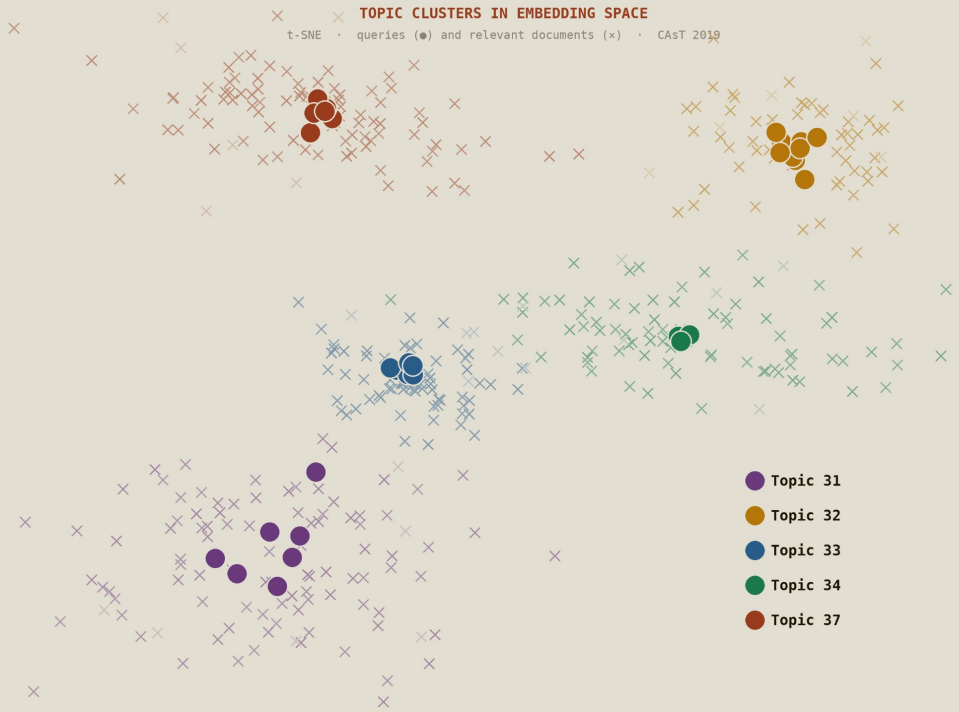
Restrict ANN search space via centroid caching or privileged entry points

Caching historical embeddings

Queries within a conversation tend to focus on related topics. In embedding space, **conversational queries and their relevant documents form tight clusters**.

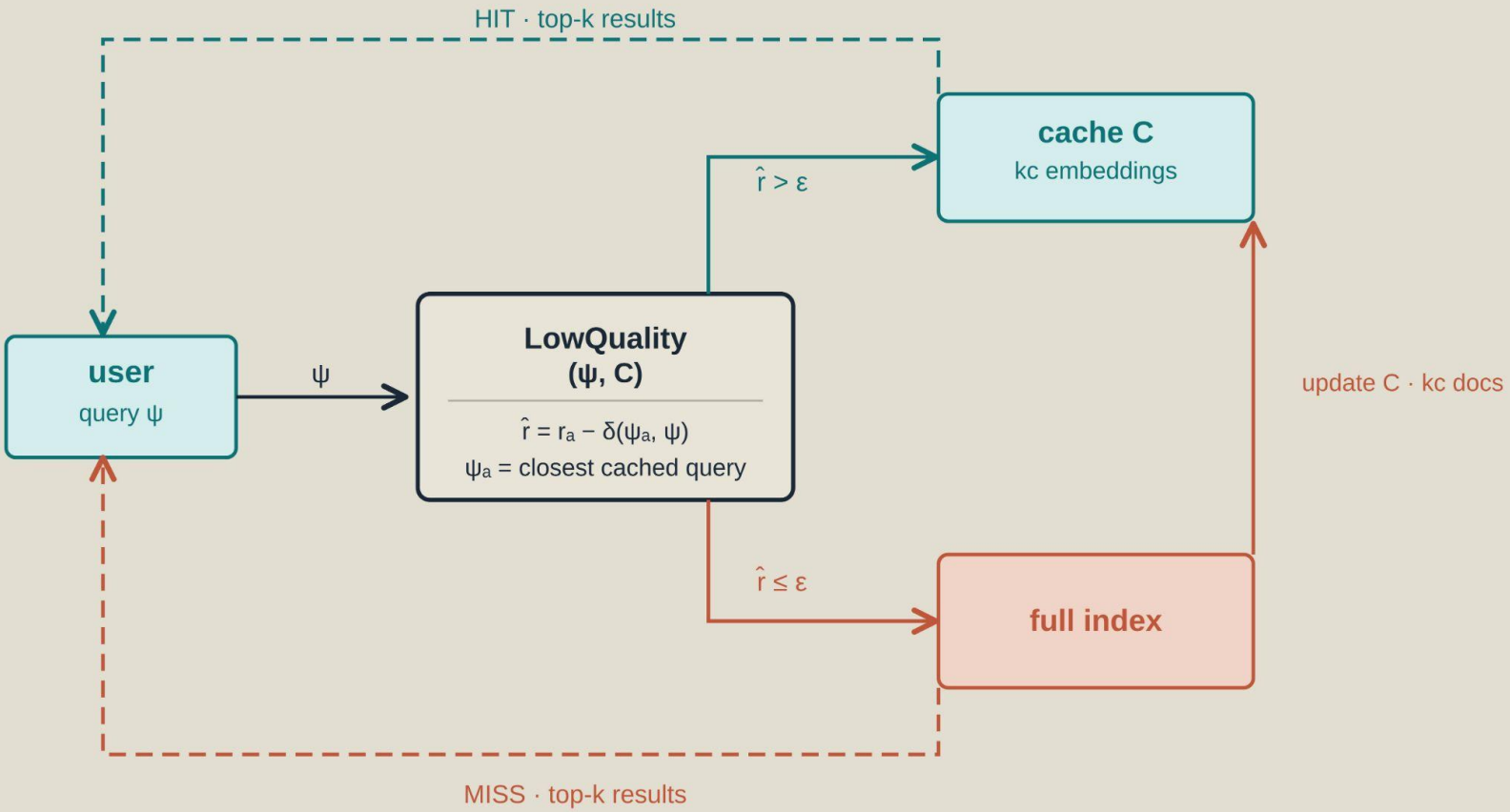
Top documents from earlier turns often reappear and enable the reuse of the candidate pool.

If we cache embeddings locally, we can answer some turns with **no** calls to the ANN backend.



Ophir Frieder, Ida Mele, Cristina Ioana Muntean, Franco Maria Nardini, Raffaele Perego, and Nicola Tonellotto. 2024. **Caching Historical Embeddings in Conversational Search**. ACM Trans. Web 18, 4, Article 42 (November 2024), 19 pages.

Client-side embedding cache

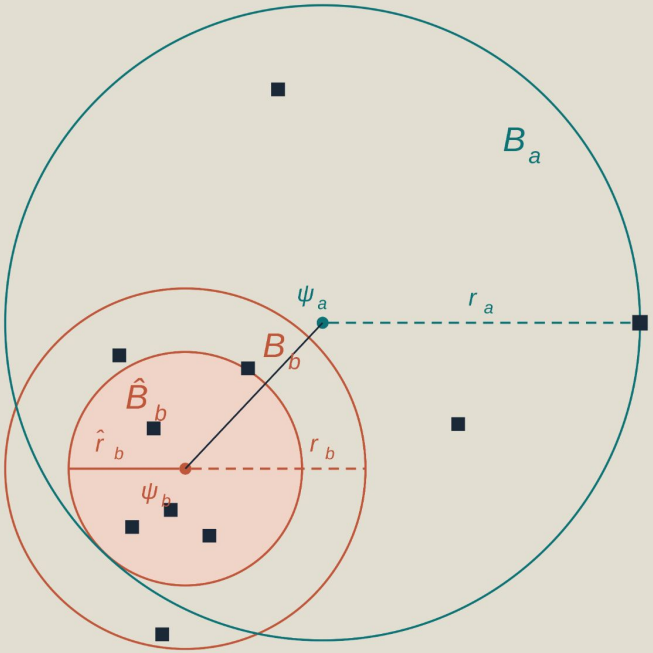


Conversational cache

Key insight is geometric:

Detect a **partial overlap** in which the query embedding ψ_b falls within the hyperball B_a

If $\hat{r}_b > 0$, we have a **guarantee** that the embeddings in B_b are the most similar to q_b among all the documents in D and the relevant documents are already in the cache.



$$\hat{r}_b = r_a - \delta(\psi_a, \psi_b)$$

Results

		k_c	MAP@200	MRR@200	nDCG@3	P@1	P@3	cov_{10}	Hit Rate
CAsT 2019	no-caching	–	**0.194**	0.647	0.376	0.497	0.495	–	–
	static-CACHE	1K	0.101▼	0.507▼	0.269▼	0.387▼	0.364▼	0.40	100%
		2K	0.112▼	0.567▼	0.304▼	0.428	0.414▼	0.47	100%
		5K	0.129▼	0.588	0.316▼	0.451	0.426▼	0.56	100%
		10K	0.140▼	0.611	0.338	0.486	0.459	0.62	100%
	dynamic-CACHE	1K	0.180▼	0.634	0.365	0.474	0.482	0.91	67.82%
		2K	0.183▼	0.631	0.366	0.480	0.487	0.93	70.69%
		5K	0.186▼	0.652	0.375	0.503	0.499	0.94	74.14%
		10K	0.190	**0.655**	**0.380**	**0.509**	**0.505**	0.96	75.29%

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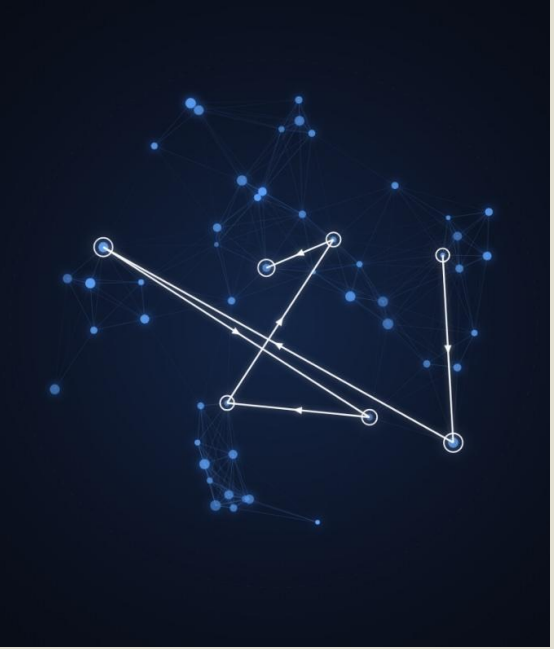


TopLoc

- ➔ Efficient conversational ANN search
- ➔ Exploits topical locality for faster response in follow-up turns
- ➔ Dynamically restrict the search space to semantically relevant documents
- ➔ Integrated into two ANN index types

IVF Inverted file index (C)

Cache "hot" centroids from the 1st query; refresh on topic shift

HNSW Graph index (G)

Set a privileged entry point from the 1st query; skip full traversal

Cristina Ioana Muntean, Franco Maria Nardini, Raffaele Perego, Guido Rocchietti, and Cosimo Rulli. 2025. **Efficient Conversational Search via Topical Locality in Dense Retrieval**. In Proc. SIGIR '25. ACM, 2749–2753.

TopLoc for inverted file index

TopLoc$_{IVF}$

- Leverages the first utterance q_0 of a conversation to identify a set C_0 of “hot” centroids
- Compares the query vector against the cached centroids in C_0

TopLoc$_{IVF+}$

- Quality measure for topic shift determines *centroid cache refresh*

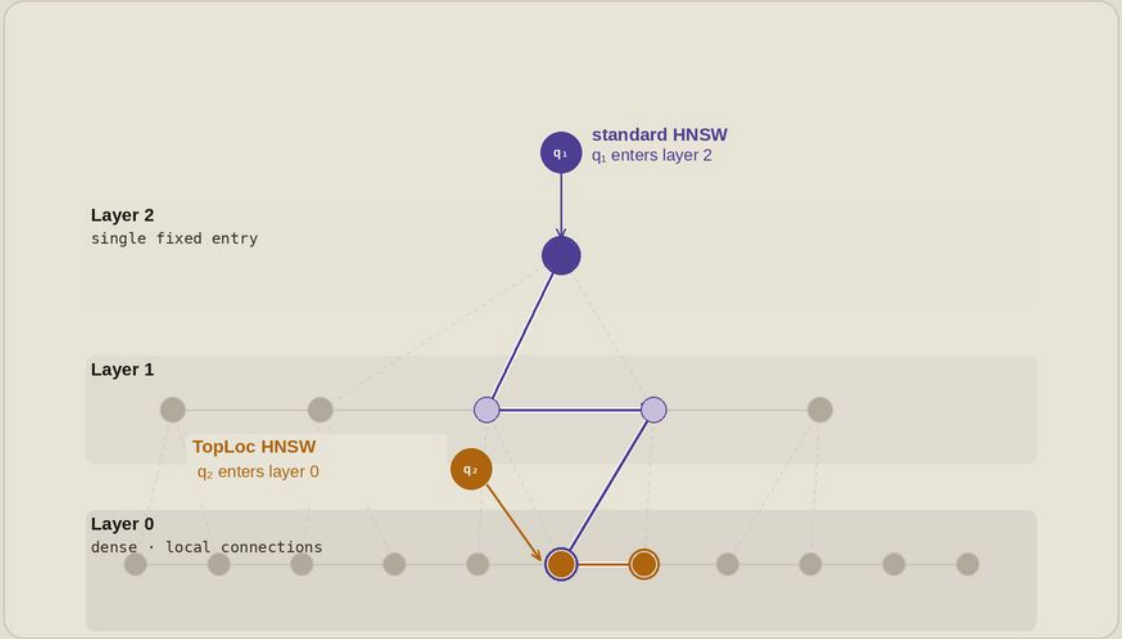
TopLoc for hierarchical graph index

$\text{TopLoc}_{\text{HNSW}}$

- Privileged entry point - the closest point to q_0

$$\text{top}_1\,(q_0, \text{D})_{HNSW}$$

- Upscaling factor $up \in [1, +\infty]$ which increases *ef-search (size of candidate list)* for the first query



Results

		TREC CAsT 2019				TREC CAsT 2020			
Model	**Search**	**MRR@10**	**NDCG@3**	**NDCG@10**	**Time**	**MRR@10**	**NDCG@3**	**NDCG@10**	**Time**
Dragon	Exact	0.799	0.522	0.492	–	0.769	0.477	0.463	–
	IVF	0.813	0.528	0.486	33.0 (–)	0.765	0.469	0.449	31.2 (–)
	TopLoc $_{\text{IVF}}$	0.789	0.517	0.479	6.5 (5.1×)	0.723	0.429	0.411	3.8 (8.2×)
	TopLoc $_{\text{IVF+}}$	0.795	0.518	0.477	3.8 (8.7×)	0.766	0.463	0.440	7.9 (3.9×)
	HNSW	0.789	0.508	0.469	8.3 (–)	0.758	0.460	0.437	7.9 (–)
	TopLoc $_{\text{HNSW}}$	0.785	0.503	0.466	0.8 (10.4×)	0.753	0.458	0.433	1.4 (5.6×)
SnowFlake	Exact	0.817	0.550	0.502	–	0.792	0.508	0.475	–
	IVF	0.815	0.544	0.497	24.9 (–)	0.785	0.507	0.462	31.4 (–)
	TopLoc $_{\text{IVF}}$	0.827	0.555	0.505	5.7 (4.4×)	0.768	0.484	0.442	6.8 (4.6×)
	TopLoc $_{\text{IVF+}}$	0.827	0.554	0.501	5.7 (4.4×)	0.781	0.502	0.456	8.1 (3.9×)
	HNSW	0.814	0.548	0.500	1.8 (–)	0.792	0.508	0.475	3.8 (–)
	TopLoc $_{\text{HNSW}}$	0.808	0.549	0.493	0.7 (2.6×)	0.791	0.507	0.475	2.2 (1.7×)

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Future insights

Conversational Search

- New evaluation benchmarks
- Task-oriented conversational search

Efficiency

- Efficiency in with generative AI solutions
- Cache in RAG
- Session efficiency

Please feel free to reach out

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